RAG-Based Customer Support Assistant (with LangGraph + HITL)

HIGH-LEVEL DESIGN

1. SYSTEM OVERVIEW

🔹 **Problem Definition**

Customer support systems require handling a large number of queries. Manual support is slow, costly, and not scalable.

🔹 **Proposed Solution**

We design a **Retrieval-Augmented Generation (RAG) based Customer Support Assistant** that:

- Reads information from a PDF knowledge base
- Retrieves relevant content
- Answers user queries automatically
- Escalates to human support when needed

🟢 2. ARCHITECTURE DIAGRAM

PDF → Loader → Chunking → Embedding → ChromaDB

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User → Query → Retriever → Answer Generator → Response

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HITL

🟢 3. COMPONENT DESCRIPTION

🔹 **Document Loader**

Loads PDF file and extracts text content.

🔹 **Chunking Module**

Splits large text into smaller chunks for better processing.

- **Embedding Model**

Converts text chunks into numerical vectors for similarity search.

- **Vector Database (ChromaDB)**

Stores embeddings and allows fast retrieval of relevant data.

- **Retriever**

Fetches most relevant chunks based on user query.

- **Answer Generator**

Generates response using retrieved content.

- **LangGraph Workflow**

Controls the flow of data and manages processing steps.

- **HITL (Human-in-the-Loop)**

Handles cases where system cannot answer confidently.

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4. DATA FLOW

1. PDF is loaded
2. Text is split into chunks
3. Chunks are converted into embeddings
4. Stored in vector database
5. User sends query
6. Retriever finds relevant chunks
7. Answer is generated
8. If no answer → escalate to human

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5. TECHNOLOGY CHOICES

- **ChromaDB** → Lightweight vector database
- **LangChain** → Framework for RAG pipeline
- **LangGraph** → Workflow control
- **HuggingFace Embeddings** → Free embedding model
- **Google Colab** → Development environment

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6. SCALABILITY CONSIDERATIONS

- Large documents handled via chunking
- Faster retrieval using vector database
- Can support multiple documents in future